

Mathematical Formulation for Decentralized UAV Swarm Navigation using Artificial Potential Fields

Autonomous UAV Swarm Control Project

Abstract

This document outlines the mathematical foundation governing the decentralized trajectory tracking and collision avoidance algorithms implemented in the ROS 2 swarm control package. The system utilizes Artificial Potential Fields (APF) to ensure mathematically provable, collision-free navigation for multiple Unmanned Aerial Vehicles (UAVs) operating in a dynamic environment.

1 Introduction

In multi-agent aerospace applications, decentralized control architectures are preferred due to their robustness and scalability. The control logic implemented relies on the continuous calculation of control forces derived from potential functions. Each UAV computes a local velocity vector based on the superposition of an attractive force towards a global target and repulsive forces away from static or dynamic obstacles.

2 Kinematic Model

Let N be the total number of UAVs in the swarm. The state of the i -th UAV is defined by its position vector $q_i \in \mathbb{R}^3$ and its velocity vector $v_i \in \mathbb{R}^3$. The single-integrator kinematic model is assumed for the high-level trajectory planner:

$$\dot{q}_i = v_i = u_i \quad (1)$$

where $u_i \in \mathbb{R}^3$ is the control input (commanded velocity) generated by the APF controller.

3 Artificial Potential Field Formulation

The total potential field $U_{total}(q_i)$ acting on the i -th UAV is the sum of the attractive potential $U_{att}(q_i)$ and the repulsive potential $U_{rep}(q_i)$:

$$U_{total}(q_i) = U_{att}(q_i) + U_{rep}(q_i) \quad (2)$$

The total desired control force F_{total} is the negative gradient of the total potential field:

$$F_{total} = -\nabla U_{total}(q_i) = F_{att} + F_{rep} \quad (3)$$

3.1 Attractive Potential

The attractive potential drives the UAV towards its designated target position q_{goal} . A quadratic potential function is utilized to ensure that the attractive force linearly decreases as the UAV approaches the target, preventing oscillations.

$$U_{att}(q_i) = \frac{1}{2}k_{att}\|q_i - q_{goal}\|^2 \quad (4)$$

where $k_{att} > 0$ is the attractive gain parameter. The resulting attractive force is:

$$F_{att} = -\nabla U_{att}(q_i) = -k_{att}(q_i - q_{goal}) \quad (5)$$

3.2 Repulsive Potential

To ensure collision avoidance, a repulsive potential field is generated around obstacles (and neighboring UAVs). The repulsive field is active only within a predefined influence radius d_0 . Let $d(q_i)$ be the shortest distance from the UAV to the obstacle.

$$U_{rep}(q_i) = \begin{cases} \frac{1}{2}k_{rep} \left(\frac{1}{d(q_i)} - \frac{1}{d_0} \right)^2 & \text{if } d(q_i) \leq d_0 \\ 0 & \text{if } d(q_i) > d_0 \end{cases} \quad (6)$$

where $k_{rep} > 0$ is the repulsive gain. The resulting repulsive force is:

$$F_{rep} = -\nabla U_{rep}(q_i) = \begin{cases} k_{rep} \left(\frac{1}{d(q_i)} - \frac{1}{d_0} \right) \frac{1}{d(q_i)^2} \nabla d(q_i) & \text{if } d(q_i) \leq d_0 \\ 0 & \text{if } d(q_i) > d_0 \end{cases} \quad (7)$$

4 Velocity Saturation and Control Output

To respect the physical aerodynamic limitations of the UAV, the calculated control force F_{total} is mapped to the commanded velocity u_i and saturated at a maximum allowable velocity v_{max} :

$$u_i = \begin{cases} F_{total} & \text{if } \|F_{total}\| \leq v_{max} \\ v_{max} \frac{F_{total}}{\|F_{total}\|} & \text{if } \|F_{total}\| > v_{max} \end{cases} \quad (8)$$

This bounded velocity vector u_i is directly translated into the `geometry_msgs/msg/Twist` linear components published to the flight controller.

5 Conclusion

By designing the control inputs as the negative gradient of a strictly positive-definite potential function, the system's stability can be analyzed via Lyapunov's direct method. The UAV is guaranteed to converge to the global minimum (the target) assuming local minima are mitigated, ensuring safe and deterministic multi-agent navigation.